Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 17-03-2026**
Time: 09.00Am to 12.00Pm **GTM-23** **Max. Marks: 300**

KEY SHEET

MATHEMATICS

1	**3**	2	**1**	3	**1**	4	**2**	5	**1**
6	**1**	7	**3**	8	**2**	9	**2**	10	**2**
11	**1**	12	**4**	13	**3**	14	**2**	15	**1**
16	**4**	17	**2**	18	**2**	19	**2**	20	**3**
21	**2**	22	**7**	23	**1**	24	**8**	25	**14**

PHYSICS

26	**4**	27	**3**	28	**3**	29	**2**	30	**4**
31	**2**	32	**2**	33	**2**	34	**1**	35	**4**
36	**2**	37	**4**	38	**1**	39	**1**	40	**2**
41	**1**	42	**2**	43	**4**	44	**2**	45	**4**
46	**1**	47	**4**	48	**3**	49	**223**	50	**40**

CHEMISTRY

51	**4**	52	**2**	53	**1**	54	**1**	55	**3**
56	**3**	57	**3**	58	**3**	59	**4**	60	**4**
61	**1**	62	**4**	63	**2**	64	**1**	65	**4**
66	**2**	67	**3**	68	**3**	69	**1**	70	**3**
71	**8**	72	**60**	73	**231**	74	**6**	75	**2**

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SOLUTION

MATHEMATICS

1. A, B, C, D are coplanar

$\left[\overline{AB}\ \overline{AC}\ \overline{AD}\right]$

$$\begin{vmatrix} -4 & -3 & 3-2\lambda \\ -7 & \lambda-5 & 4-2\lambda \\ -6 & 0 & 6-2\lambda \end{vmatrix} = 0$$

$\lambda^2 - 5\lambda + 6 = 0$

$(\lambda-3)(\lambda-2) = 0$

$Now \sum_{\lambda \in S} (\lambda+3)^2 = 36 + 25 = 61$

2. Probability$= \left(3_{C_2}\right)\left(\frac{1}{9}\right)^2\left(\frac{8}{9}\right) = \frac{8}{243}$

3. $A^2 = I \Rightarrow |A^2| = 1 \Rightarrow |A| = \pm 1 = b$

$A = \begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}$

$A^2 = I$

$\begin{bmatrix} \alpha^2+\beta\gamma & \alpha\beta+\beta\delta \\ \alpha\gamma+\gamma\delta & \gamma\beta+\delta^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Let $\alpha^2 + \beta\gamma = 1$

$(\alpha+\delta)\beta = 0 \Rightarrow \alpha+\delta = 0 = a$

$(\alpha+\delta)\gamma = 0$

$\beta\gamma + \delta^2 = 0$

4. $\frac{n_{C_r}}{n_{C_{r-1}}} = 5 \qquad \frac{n_{C_{r+1}}}{n_{C_r}} = 4$

$\frac{n-r+1}{r} = 5 \qquad n = 5r+4 \rightarrow (2)$

$n = 6r - 1 \rightarrow (1)$

$n = 29r = 5$

$Coeff\ of\ 4th\ term = 29_{C_3} \quad = 3654$

5. $A^{31}B^{52} - B^{52}A^{31}$ is skew symmetric

$A^{52}C^{31} - C^{31}A^{52}$ is symmetric

6. $4\cos^2\theta - 1 = 4(1-\sin^2\theta) - 1 = 3 - 4\sin^2\theta \quad = \frac{\sin 3\theta}{\sin\theta} \rightarrow (1)$

So, given expression can be written as

$$= 54 \times \frac{27^0}{\sin 9^0} \times \frac{\sin 81^0}{\sin^{27^0}} \times \frac{\sin 279^0}{\sin 243^0} [\because By\ (1)] = 54 \times \frac{\sin 729^0}{\sin 9^0} = 54$$

7. $a^n + b^n$ is divisible by a + b only if n is odd.

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8. $([x]+3)([x]-6)>0$

9. Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

Given Re($z_1 z_2$)=0 $\quad x_1x_2 - y_1y_2 = 0$

So, x_1 and x_2 are of opposite sign and y_1, y_2 are of opposite sign.

10. $f(x) = x^3 - 3ax^2 + 3(a^2-1)x + 1$

$f^1(x) = 3x^2 - 6ax + 3(a^2-1)$

$= 3(x-(a+1))(x-(a-1))$

$f^1(x) = 0 \Rightarrow x = a+1 \text{ or } a-1$

So, $\therefore a+1 \in (-2,4)$ *if* $a \in (-3,3)$

and $a-1 \in (-2,4)$ *if* $a \in (-1,5)$

$\therefore a \in (-1,3)$

11. $A------ = 4!$

$H------- = 4!$

$M------- = 4!$

$S------- = 4!$

$TA------ = 3!$

$THAMS = 1$

$Total = 4\times 4! + 1\times 3! + 1 = 103$

12. We have $f(x) = \sqrt{-\sin^{-1}(\sqrt{\log_4^x}} + \sin^{-1}\left(\frac{1+x^2}{4x}\right)$

Clearly domain of $f(x)$ is x=1 only

So $f(1) = \sqrt{0} + \sin^{-1}\left(\frac{2}{4}\right) = \frac{\pi}{6}$

Hence Range of $f(x) = \left\{\frac{\pi}{6}\right\}$

13. $y = (C_1 \operatorname{Cos} C_2)\operatorname{Cos} x + (C_5 - C_1 \operatorname{Sin} C_2)\operatorname{Sin} x - C_3 e^{C_4} e^{-x}$

$y = A\operatorname{Cos} x + B\operatorname{Sin} x + Ce^{-x}$ $\quad \because C_1, C_2, C_3, C_4$ are arbitrary constants.

14. $RHL = \underset{x\to 0^+}{Lt} \frac{\operatorname{Cos}^{-1}(1-x^2)\operatorname{Sin}^{-1}(1-x)}{x(1-x^2)} = \frac{\pi}{2} \underset{x\to 0^+}{Lt} \frac{\operatorname{Cos}^{-1}(1-x^2)}{x}$

$= \frac{\pi}{2} \underset{x\to 0^+}{Lt} \frac{-1}{\sqrt{1-(1-x^2)}}(-2x)$ $\left(L^1 \text{ Hospital Rule}\right)$

$= \pi \underset{x\to 0^+}{Lt} \frac{x}{\sqrt{2x^2 - x^4}} = \pi \underset{x\to 0^+}{Lt} \frac{1}{\sqrt{2-x^2}} = \frac{\pi}{\sqrt{2}}$

$LHL = \underset{x\to 0^+}{Lt} \frac{\operatorname{Cos}^{-1}(1-(1+x^2)\operatorname{Sin}^{-1}(-x)}{(1+x)-(1+x)^3}$

$= \frac{\pi}{2} \underset{x\to 0^+}{Lt} \frac{\operatorname{Sin}^{-1} x}{(1+x)\left[(1+x)^2 - 1\right]} = \frac{\pi}{2} \underset{x\to 0^-}{Lt} \frac{\operatorname{Sin}^{-1} x}{x^2+2x} = \frac{\pi}{2}\left(\frac{1}{2}\right) = \frac{\pi}{4}$

As $LHL \neq RHL$

So $f(x)$ *is not continuous at* $x = 0$ $\quad \therefore$ *No Such* α *exists*

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15. $\int_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}} \frac{48}{\sqrt{3^2-(2x)^2}}dx = 48.\frac{1}{2}\text{Sin}^{-1}\left(\frac{2x}{3}\right)\Bigg|_{\frac{3\sqrt{2}}{4}}^{\frac{3\sqrt{3}}{4}}$

16. $C_1 \to C_1 + C_2$

$$f(x) = \begin{vmatrix} 2 & 1+\text{Cos}^2 & \text{Cos}\,2x \\ 2 & \text{Cos}^2 x & \text{Cos}\,2x \\ 1 & \text{Cos}^2 x & \text{Sin}\,2x \end{vmatrix}$$

$R_1 \to R_1 - R_2$

$$f(x) = \begin{vmatrix} 0 & 1 & 0 \\ 2 & \text{Cos}^2 x & \text{Cos}\,2x \\ 1 & \text{Cos}^2 x & \text{Sin}\,2x \end{vmatrix}$$

$f(x) = -2\,\text{Sin}\,2x + \text{Cos}\,2x$

$f(x)_{\max} = \sqrt{5}$

17. $x^2 + 5^{\frac{1}{2}} = -(20)^{\frac{1}{4}} \Rightarrow \left(x^2+\sqrt{5}\right)^2 = \sqrt{20}\,x^2$

$\Rightarrow x^4 + 5 + 2\sqrt{5}x^2 = 2\sqrt{5}x^2 \Rightarrow x^4 = -5 \Rightarrow x^8 = 25 \qquad So; \alpha^8 + \beta^8 = 50$

18. Any point $(2\lambda+1,\ 3\lambda-2,\ 6\lambda+3)$

Given $4\lambda^2 + 9\lambda^2 + 36\lambda^2 = 9$

$49\lambda^2 = 9,\ \lambda = \pm\frac{3}{7}, g(x) = \sqrt{x}+1$

19. $\frac{dy}{dx} = 1 + xe^{y-x}$

$e^{-y} = t \Rightarrow e^{-y}\frac{dy}{dx} = \frac{dt}{dx}$

$-\frac{dt}{dx} = t + xe^{-x}$

$\frac{dt}{dx} + t = -xe^{-x}$

$\therefore Integrating\ \ factor = e^{\int 1.dx} = e^x$

$\therefore Solution\ is\ t.e^x = \int -x\,e^{-x}\,e^x\,dx$

$t.e^x = -\frac{x^2}{2} + C \Rightarrow e^{x-y} = \frac{-x^2}{2} + C$

$\because y(0) = 0 \Rightarrow C = 1$

$\therefore e^{x-y} = 1 - \frac{x^2}{2} \Rightarrow y(x) = x - \ln\left(1 - \frac{x^2}{2}\right)$

20. A (1, 0, 7); B (1, 6, 3)

Mid-Point of AB = (1, 3, 5) lies in the line

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DR'S of AB (0, 6, -4) the line passing through A and B is perpendicular to the given line

Hence B is the mirror image.

Statement II is also true but not the correct explanation of I as there are infinitely many lines passing through the midpoint of line segment and one of the lines is perpendicular bisects.

21. Area bounded $= \int_{1/e}^{1} \frac{\log x}{ex}$

$$= \frac{1}{e}\left[\frac{(\log x)^2}{2}\right]_{1/e}^{1} - e\left(\frac{x^2}{2}\log x - \frac{x^2}{4}\right)_{1/e}^{1} = \frac{1}{e}\left(-\frac{1}{2}\right) - e\left(\frac{2}{2e^2} - \frac{1-1/e^2}{4}\right)$$

$$= -\frac{1}{2e} - \frac{1}{2e} + \frac{e}{4} - \frac{1}{4e} = \frac{e}{4} - \frac{5}{4e}$$

22. $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 5 & \lambda \end{vmatrix} = 0 \qquad D_3 = \begin{vmatrix} 1 & 1 & 16 \\ 1 & 2 & 14 \\ 2 & 5 & \mu \end{vmatrix} = 0$

$\lambda - 8 = 0 \qquad \mu - 36 = 0$

$\lambda = 8 \qquad \mu = 36$

23. The Coefficient of x^9 in $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$ is $11_{C_6}\frac{\alpha^5}{\beta^6}$

The Coefficient of x^{-9} in $\left(\alpha x + \frac{1}{\beta x^3}\right)^{11}$ is $-11_{C_5}\frac{\alpha^6}{\beta^5}$ $\qquad (\alpha\beta)^2 = 1$

24. For ellipse $4 = 18(1-e^2_1)$ $\qquad e^2_1 = \frac{7}{9} \Rightarrow e_1 = \frac{\sqrt{7}}{3}$

For hyperbola $4 = 9\left(e^2_2 - 1\right)$ $\qquad e_2 = \frac{\sqrt{13}}{3}$

Put (e_1, e_2) in $15x^2 + 3y^2 = K$ $\qquad \therefore K = 16 \qquad \therefore \frac{K}{2} = 8$

25. Lines are $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ $\qquad \frac{x-6}{3} = \frac{y-1}{-2} = \frac{z+8}{0}$

$$= \frac{\left[\vec{c}-\vec{a}\ \ \vec{b}\ \ \vec{d}\right]}{\left|\vec{b}\times\vec{d}\right|}, \vec{b} = 3\hat{i} + 2\hat{j} + 2\hat{k}, \vec{d} = 3\hat{i} - 2\hat{j}, \vec{a} = 2\hat{i} - \hat{j} + 6\hat{k}, \vec{c} = 6\hat{i} + \hat{j} - 8\hat{k}$$

$$S.D = \frac{\begin{vmatrix} 4 & 2 & -14 \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}} = \frac{6+12+168}{\left|-4\hat{i} + 6\hat{j} - 12\hat{k}\right|} = \frac{196}{14} = 14$$

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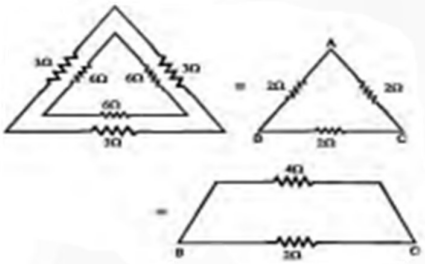
PHYSICS

26. Let the scooterist velocity be v. Then

$$1000+(10\times100)=v\times100\Rightarrow100v=2000\Rightarrow v=\frac{2000}{100}=20m/s$$

27. $$\theta=\frac{K_1\phi_1l_2+K_2\phi_2l_1}{K_1l_2+K_2l_1}=\frac{K\times0\times2+3K\times100\times1}{K\times2+3K\times1}=60^{\circ}C$$

28. $$R_{eq}=\frac{R_1R_2}{R_1+R_2}(Parallel)=\frac{4\times2}{4+2}=\frac{8}{6}=\frac{4}{3}\Omega$$



29. $$\frac{q(Q_1-Q_2)\left(\sqrt{2}-1\right)}{\sqrt{2}4\pi\varepsilon_0R}$$

30. $$T^1=2\pi\sqrt{\frac{l}{g_{eff}}}=2\pi\sqrt{\frac{l}{3g/2}}\sqrt{\frac{2}{3}}T$$

31. $$x_{cm}=\frac{\int_0^L(dm)x}{\int_0^L dm}=\frac{\int_0^L\left(a+\frac{bx}{L}\right)dx.x}{\int_a^L\left(a+\frac{bx}{L}\right)dx}$$

32. $$C_V=\frac{fR}{2}$$

$$C_P-C_V=R$$

$$C_P=R+CV=R+\frac{fR}{2}=R\left(1+\frac{f}{2}\right),\qquad\gamma=\frac{C_P}{C_V}$$

33. $$I=\frac{9mR^2}{2}-\left[\frac{mR^2}{18}+\frac{4mR^2}{9}\right]=\frac{9mR^2}{2}-\left[\frac{mR^2}{9}\left[\frac{1}{2}+4\right]\right]$$

$$=\frac{9mR^2}{2}-\left[\frac{mR^2}{9}\left[\frac{9}{2}\right]\right]=\frac{9mR^2}{2}-\frac{mR^2}{2}=\frac{8mR^2}{2}=4mR^2$$

34. $$\beta_{\max}=\frac{\lambda D}{d_0-a_0}\qquad\beta_{\min}=\frac{\lambda D}{d_0+a_0}$$

35. $$\bar{E}=-9\sin\left[\left(200\pi\left(y+ct\right)\right)\right]\hat{k}\,v/m$$

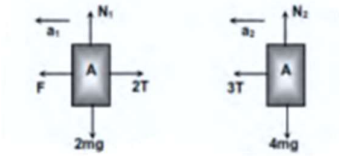
36. For an adiabatic process,

$$0=dU+PdV=0\ \ or\ \ d(a+bPV)+PdV=0\ \ \ or\ \ (b+1)\frac{dV}{V}+b\frac{dP}{P}=0$$

$$or\ \ (b+1)\log V+b\log P=cons\tan t$$

$$V^{b+1}p^b=cons\tan t\ \ \ or\ PV^{\frac{b+1}{b}}=cons\tan t\qquad\therefore\gamma=\frac{b+1}{b}$$

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37. 

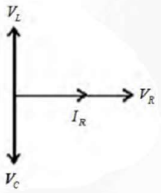
$F-2T=2ma_1$

$3T=4ma_2$

$2a_1=3a_2$

solving we get $a_2=\frac{3F}{17m}$

38. Both A and R are true but R is the correct explanation of A

39. 

A) Phase difference between current and voltage in purely resistive AC circuit is zero.

B) Phase difference between current and voltage in a pure inductive AC circuit is $\frac{\pi}{2}$; current lags voltage.

C) Phase difference between current and voltage in a pure capacitive AC circuit is $\frac{\pi}{2}$; current leads voltage.

D) Phase difference between current and voltage in an LCR series circuit is $\tan^{-1}\left(\frac{X_C-X_L}{R}\right)$.

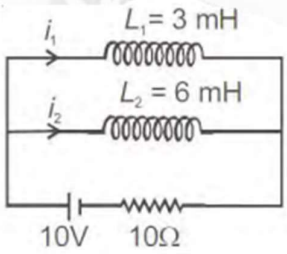
40. $C=\frac{\in_0 A}{d}$ & $C=2C_1+2C_2+C_3$

41. At node, energy is maximum when all particle reach to there extreme position.

42. A – S; B – P; C – Q; D – R

43. Statement – 1 is False, Statement –2 is True.

44. As magnetic flux is same through inductors, $L_1i_1=L_2i_2$



$\Rightarrow i_1=2i_2\ldots..(i)$

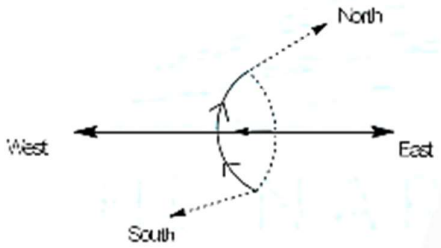
$Also, i_1+i_2=\frac{\varepsilon}{R}=\frac{10V}{10\Omega}=1A\ldots..(ii)$

$(i)\,and\,(ii)$

$\Rightarrow i_1=\frac{2}{3}A$

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45.



$$B=\frac{\mu_0 i}{2\pi d}$$

$$=\frac{\left(4\pi\times10^{-7}\right)\times(100)}{2\times3.14\times4}$$

$$B=5\times10^{-6}T\ (Southwards)$$

46. Given $v_0=\frac{v_e}{2}$ $\sqrt{\frac{GM}{R+h}}=\frac{1}{2}\sqrt{\frac{2GM}{R}}$

$$h=R$$

When the satellite stops, K=0 $-\frac{GMm}{2R}-\left(-\frac{GMm}{R}\right)=\frac{1}{2}mv^2$

$$v=\sqrt{gR}$$

47. Use $\mu=\frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin(A/2)}$

$$\delta m=A$$

48. $\lambda=\frac{h}{\sqrt{2mqv}}$

$$\frac{\lambda_p}{\lambda_a}=\frac{\sqrt{m_a q_a}}{\sqrt{m_p q_p}}=\sqrt{8}=2\sqrt{2}=3$$

49. L.C.=0.01 mm zero error =-0.08 mm

Reading $=2$ mm $+15\times0.01+0.08=2.23$ mm

50. Using the principal of calorimetry

$$M_{ice}L_f+m_{ice}(40-0)C_w=m_{stream}L_v+m_{stream}(100-40)C_w$$

$$\Rightarrow M(540)+m\times1\times(100-40)=200\times80+200\times1\times40$$

$$\Rightarrow 600M=24000$$

$$\Rightarrow M=40g$$

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CHEMISTRY

51. P: Markonikoff product with rearrangement
Q: Antimarkonikoff product
R: Markonikoff product without rearrangement

52. Bond dissociation Enthalpy $cl_2 > Br_2 > F_2 > I_2$

53. $\Delta P.E = -2\left[K.E_f - K.E_i\right]$

54. R=Phenol, S = Acalione

55. Lassaigne's test for nitrogen is given by those compounds in which N is bonded to carbon.

56. PCl3F2 has zero dipole moment

57. Even though Ce+4 is favoured by its noble gas configuration, it is strong oxidant, reverting to common oxidation state of +3. E0 of Ce4+/Ce+3= 1.74V suggests that Ce4+ can oxidize even water(but reaction is slow)

58. $[Co(NH_3)_2(NO_2)]^+$ and $[Co(NH_3)_4(NO_2)(ONO)]^+$ are linkage isomers.
$[Co(NH_3)_2(NO_2)]^+$ exhibits geometrical isomers but both the geometrical isomers are optically inactive
$[Co(NH_3)_2(NO_2)]Cl$ and $[Co(NH_3)_2(NO_2)Cl]NO_2$ are ionisation isomers.

59. $Sb_2S_3 \rightleftarrows 2Sb^{3+} + 3S^{2-}$
Eq. Conc. 2S 3S (Suppose solubility of Sb_2S_3 is S moles L^{-1})
$K_{sp} = (2s)^2(3s)^3 = 108\times(10^{-5})^5 \Rightarrow K_{sp} = 108\times10^{-25}$

60. For isotonic sol $\pi_1 = \pi_2$
$8.21 = C\times0.0821\times310$
$C = 0.323$ mol/lit ; wt of glucose $= 0.323\times180 = 58.14$

61. $\left[H^+\right] = \sqrt{K_a\times C}$ $p^H = -\log\left[H^+\right]$

62. Intra molecular Eserfication

63. S^1N favourable condition

64. 3 L.P

65. But maximum yield is para(51%)

66. $O_2^{2-}(BO=1)O_2^-(BO=1.5)O_2(BO=2)O_2^+(BO=2.5)$

67. 5.6 lit of O_2 liberated

68. No. of angular nodes=l, No. of radial (spherical) nodes=$n-l-1$
The no of peaks in radial probability distribution curves=$n-l$

69. $\left[Fe(CN)_6\right]^{4-}$ and $\left[Fe(H_2O)_6\right]^{3+}$ have same number of unpaired e^- in central metal ion

70. Inert pair effect

71. Number of $-OH$ groups in one molecule of sucrose is 8

72. $-180 = -RE - 240$ $RE = 180 - 240 = 60 Kcal\, mol^{-1}$

73. $k = \dfrac{Rate}{[A]} = 3\times10^{-3} = \dfrac{0.693}{t_{1/2}}$

74. number of molecules with sp^3d hybridization is 6

75. $\Delta n = 0,\ K_p = K_c$

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